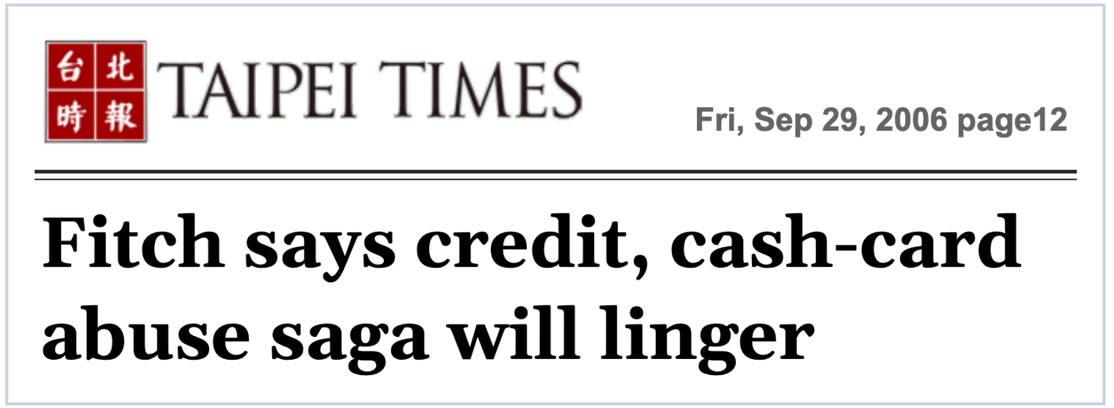
Card default: the cut-off matters more than the model

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Ten classifiers compared on training CV; three ensembles tie; the frozen test set only reports.

RUAN Hangjing · G2607546F
LU Xinyu · G2603537D
LI Jingwen · G2607526K

MH6805 Machine Learning in Finance
Group B19 · 27 October 2026



台北時報 TAIPEI TIMES

Fri, Sep 29, 2006 page12

Fitch says credit, cash-card abuse saga will linger

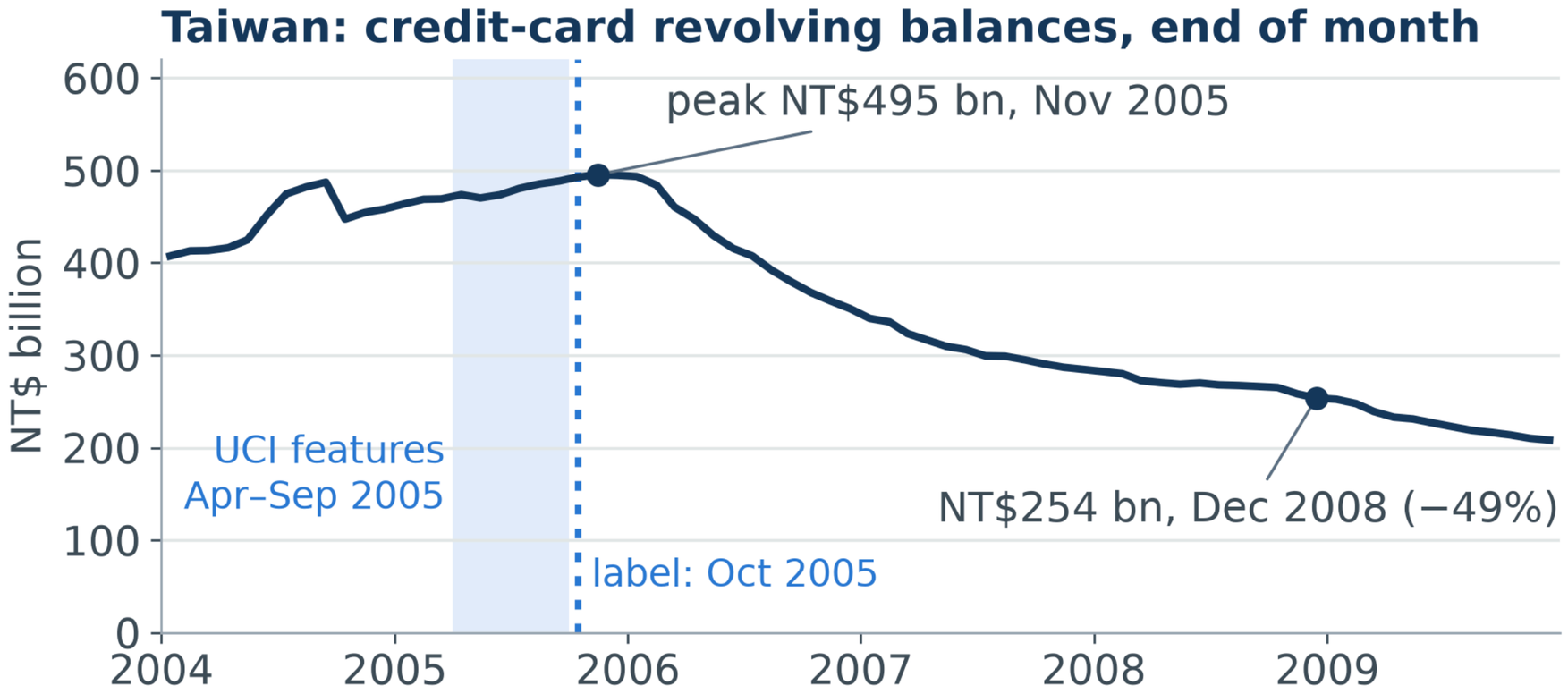
Taipei Times, 29 Sep 2006: the crisis our data come from

Contribution (draft, to confirm): RUAN 29% — KNN benchmark, logistic family, clustering, nested-CV pass · LU 29% — tree, forest, boosting, SHAP/PDP/monotone · LI 42% — SVM/MLP/discriminants, imbalance, leakage, protocol and freeze, cost analysis, deck. Per-section table in the report.
Presenting: RUAN 1–7 · LU 8–13 · LI 14–21

Taipei Times, 29 Sep 2006, p. 12, https://www.taipeitimes.com/News/biz/archives/2006/09/29/2003329739 · UCI Default of Credit Card Clients (Yeh & Lien 2009), doi:10.24432/C55S3H, CC BY 4.0 · results/aggregate_summary.json

Our data come from Taiwan's 2005 card-debt crisis

29,965 cardholders; 22.1% default next month; "nobody defaults" is 77.9% right, catches none.

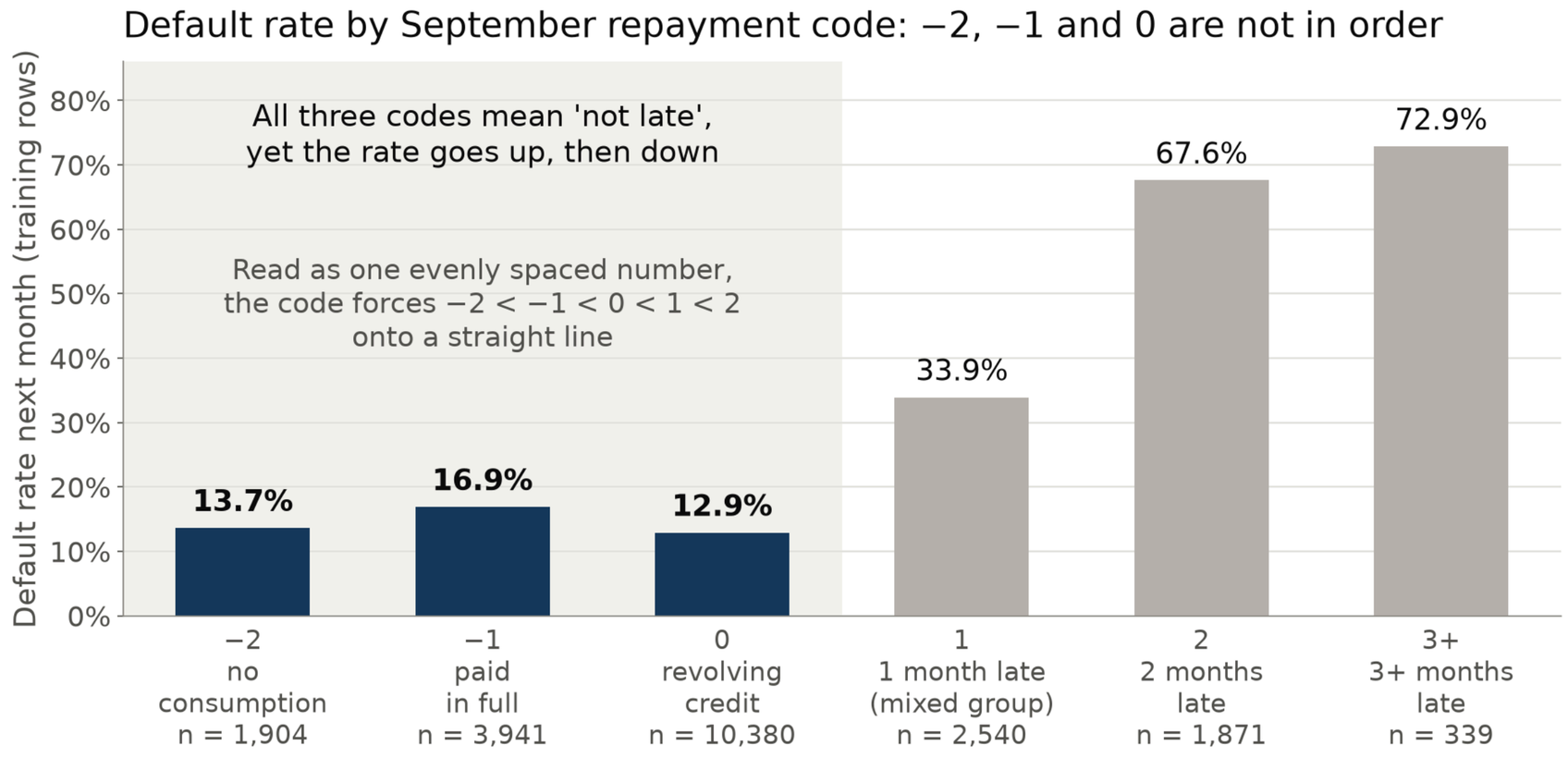


Taiwan's card revolving balances peaked in Nov 2005, then fell 49% by 2008

FSC Banking Bureau, credit-card statistics (index-7.xlsx, OGDL v1.0), https://www.banking.gov.tw/webdowndoc?file=/stat/cc/index-7.xlsx · CBC Financial Stability Report, Jun 2008, Box 2, https://www.cbc.gov.tw/en/public/Attachment/931716211071.pdf · results/eda_summary.json

Repayment codes are states, not a number line

Two months late: 67.6% default; “paid in full” defaults more than “revolving” (16.9% vs 12.9%).



September default rate by status code, training rows: −2, −1 and 0 have no order

Training rows only (n = 20,975); results/pay_code_default_rates.csv, eda_summary.json; codes −2 and 0 are not defined in the UCI codebook

Every choice made on training folds; test only reports

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Five-fold training CV picks settings and cut-offs; each model frozen at its first test scoring.

29,965 clients
23 raw + 6 engineered features

→

ONE stratified 70/30 split
seed 2607526
saved to disk

→

Pipeline inside 5-fold CV
scale · encode · resample · fit
re-fitted in every fold

→

Out-of-fold probabilities
rank by CV AUC (OOF for SVM)
thresholds: F1 and 5:1 cost

→

FREEZE
manifest of 46 configurations
guard refuses any change

TRAINING ROWS DECIDE EVERYTHING (20,975)

round 1 scored 16–18 Sep, registered 19 Oct · 2 re-fits 30 Sep · 9 Platt 19 Oct = 46 entries

Anything chosen afterwards (recalibration maps, cut-offs, cluster labels) is chosen on training out-of-fold scores only.

Nested CV (30 Sep, RUAN):
almost the same AUC; its rule re-selects the frozen KNN and logit settings

Post-freeze checks (21 Oct): training CV only; select nothing

TEST SET 8,990 CLIENTS — EACH CONFIGURATION FROZEN AT FIRST SCORING · EVERY READ DATED

test AUC · AP · Brier · cost at the training-chosen cut-off are reported, never used to rank, tune or select

bootstrap intervals and cost curves read test rows but select nothing



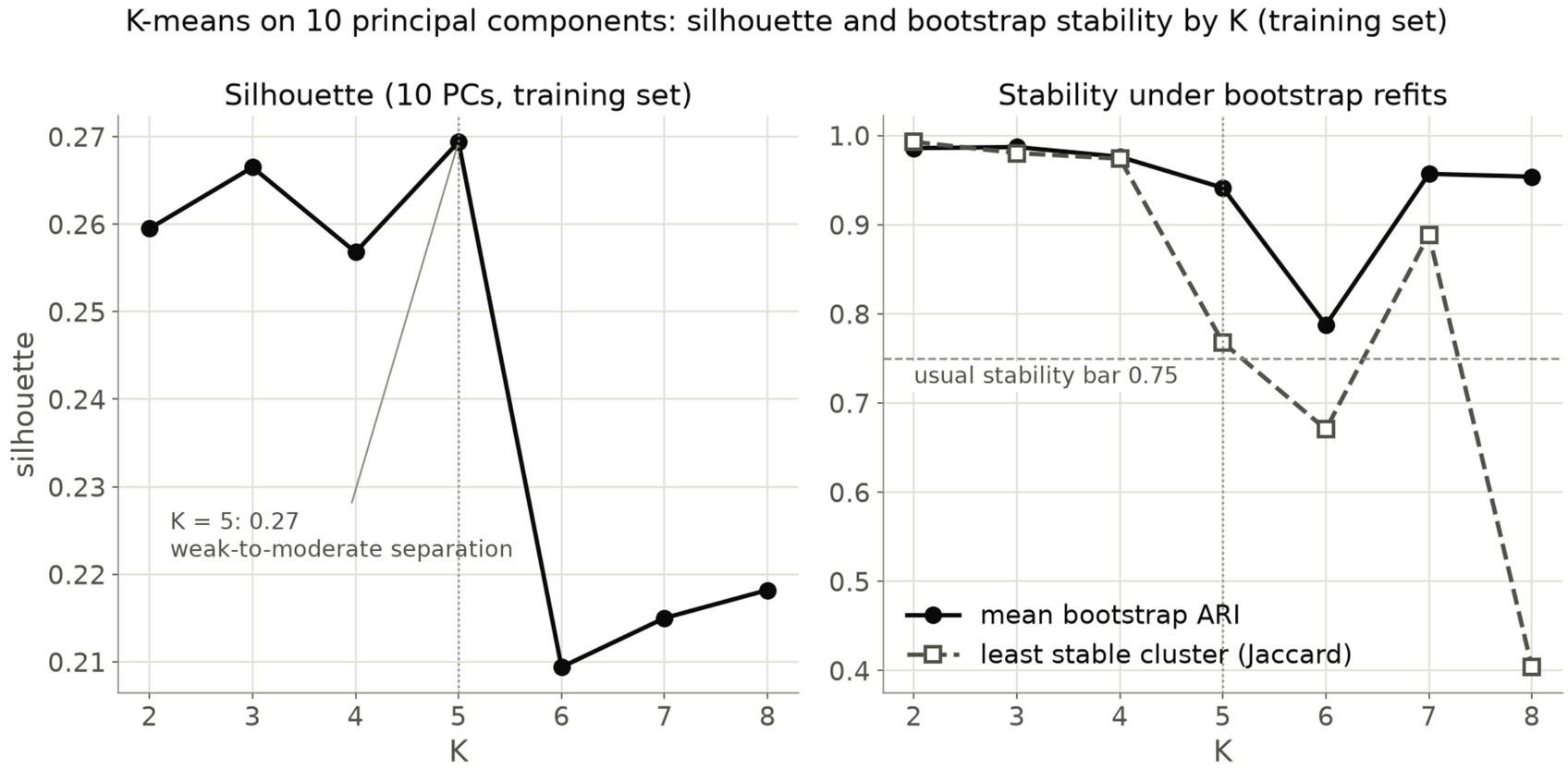
register since 19 Oct; earlier reads dated in TEST_SET_LOG.md

Rule: models ranked by training CV AUC, cut-offs from training out-of-fold scores; a changed configuration is refused before any test label is read.

results/frozen_manifest.json (46 entries), TEST_SET_LOG.md (every test read, dated), nested_cv_summary_repro.json; code/freeze.py --check passes (freeze machinery: LI)

Segments describe the book; they do not price risk

K-means finds five overlapping segments (silhouette 0.27); the label adds no measurable AUC.

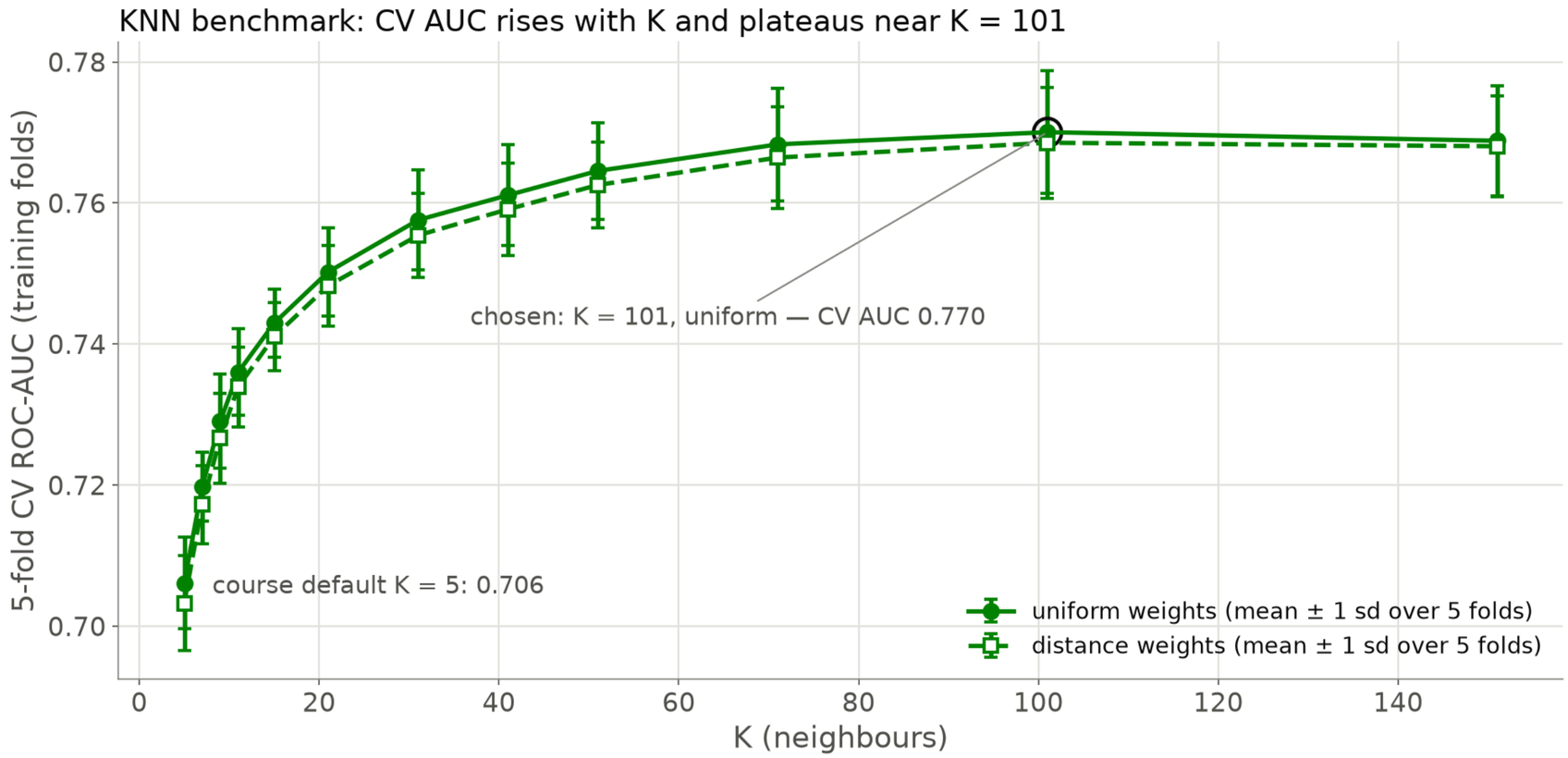


Silhouette (left) and bootstrap stability (right) by K, training rows; dashed line = usual stability bar 0.75

Training rows only (RUAN's 30 Sep analysis, reproduced 21 Oct); results/unsupervised_summary_repro.json, kmeans_selection_repro.csv, cluster_incremental_folds_repro.csv

The KNN benchmark needs about a hundred neighbours

K = 101 reaches CV AUC 0.770; nested CV, re-tuning inside each fold, gives 0.768.

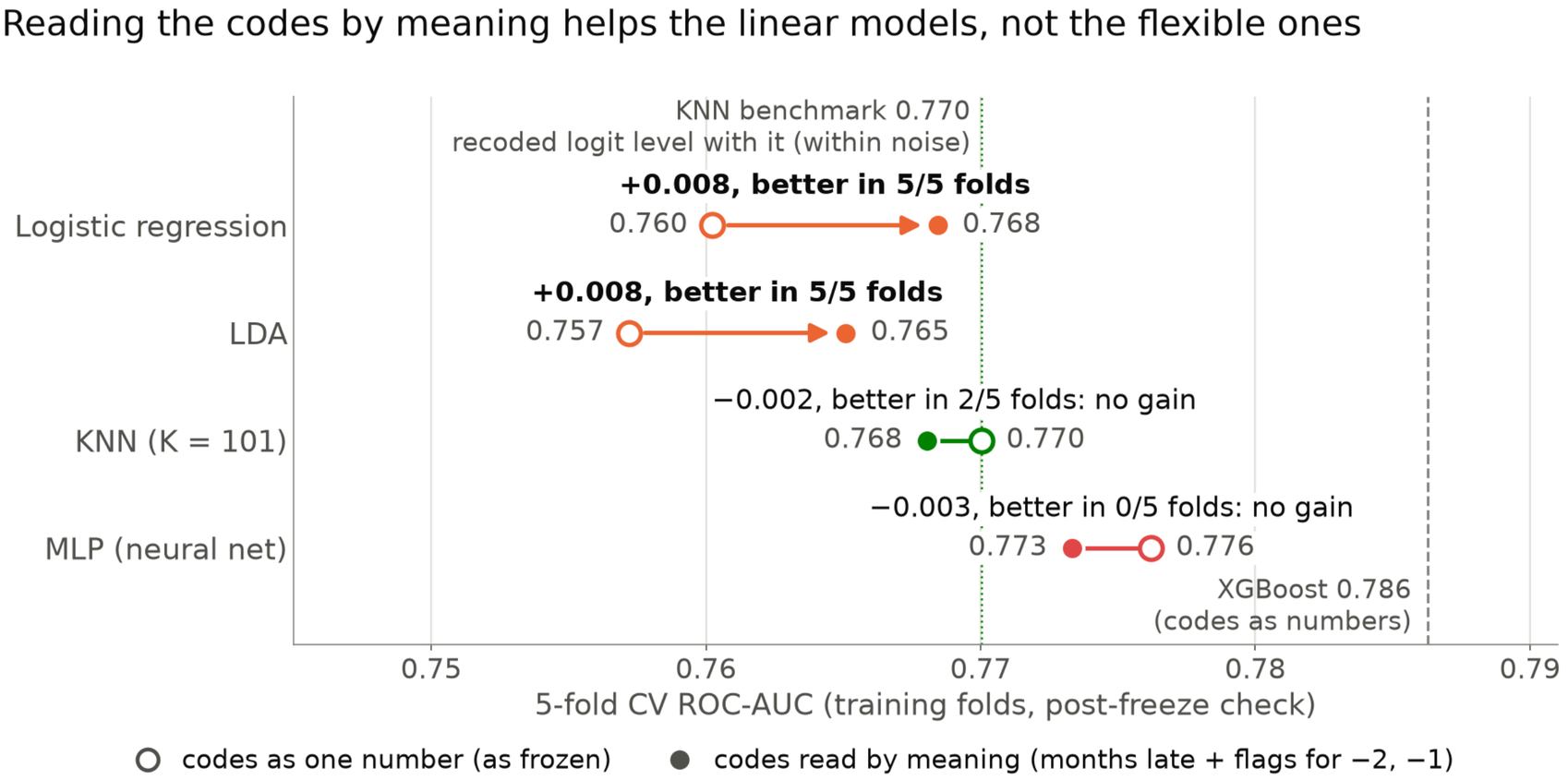


CV AUC rises with K and levels off near K = 101 (training folds)

Training 5-fold CV; results/knn.json, knn_k5.json, nested_cv_knn_frozen_grid.json, knn_dimensionality_cv.csv

Why the logit trails: coding first, shape second

Coding status by meaning lifts the logit from 0.760 to 0.768, level with KNN (training CV).



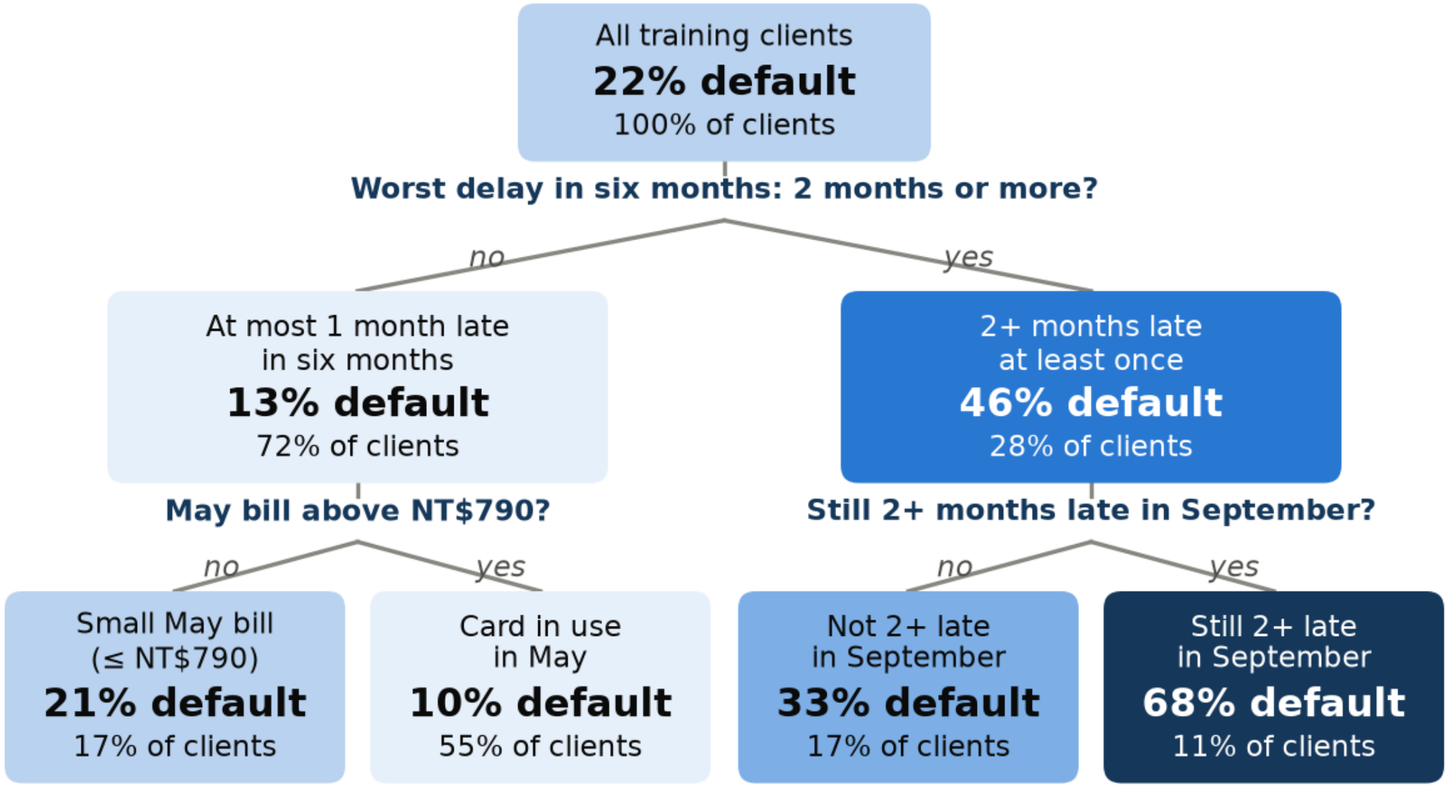
CV AUC, codes as numbers vs read by meaning: only the linear models gain

Post-freeze training CV, never test-scored; results/pay_coding_cv.csv, pay_coding_summary.json

Pruned tree ties KNN; ensembles tie at the top

Forest, gradient boosting and XGBoost reach CV AUC 0.785–0.786, about 0.025 above the logit.

One pruned tree reads like a credit policy (top 2 of 5 levels; 17 rules in all)



ONE TREE

17 rules, first split on worst delay; ties KNN (0.769)

RANDOM FOREST

Averages 500 trees grown on resampled clients: errors cancel

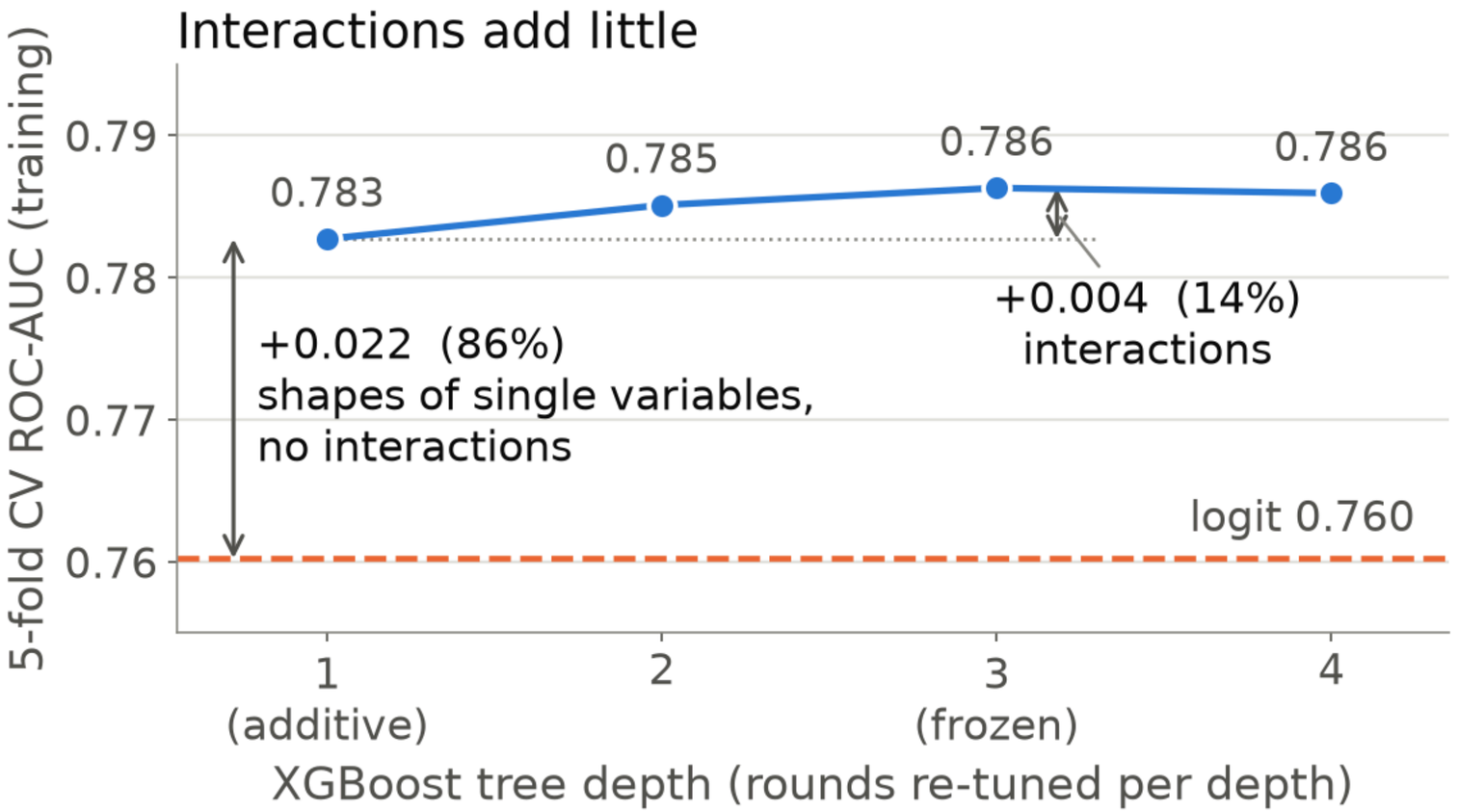
BOOSTING

Adds small trees, each fixing the errors left

Training 5-fold CV; results/tree_pruned.json, tree_rules.txt, random_forest.json, gradient_boosting.json, xgboost.json, fold_stability.csv

XGBoost’s edge is mostly shape, not interactions

One-split trees recover 86% of XGBoost’s lead; interactions add only 0.004 (training CV).

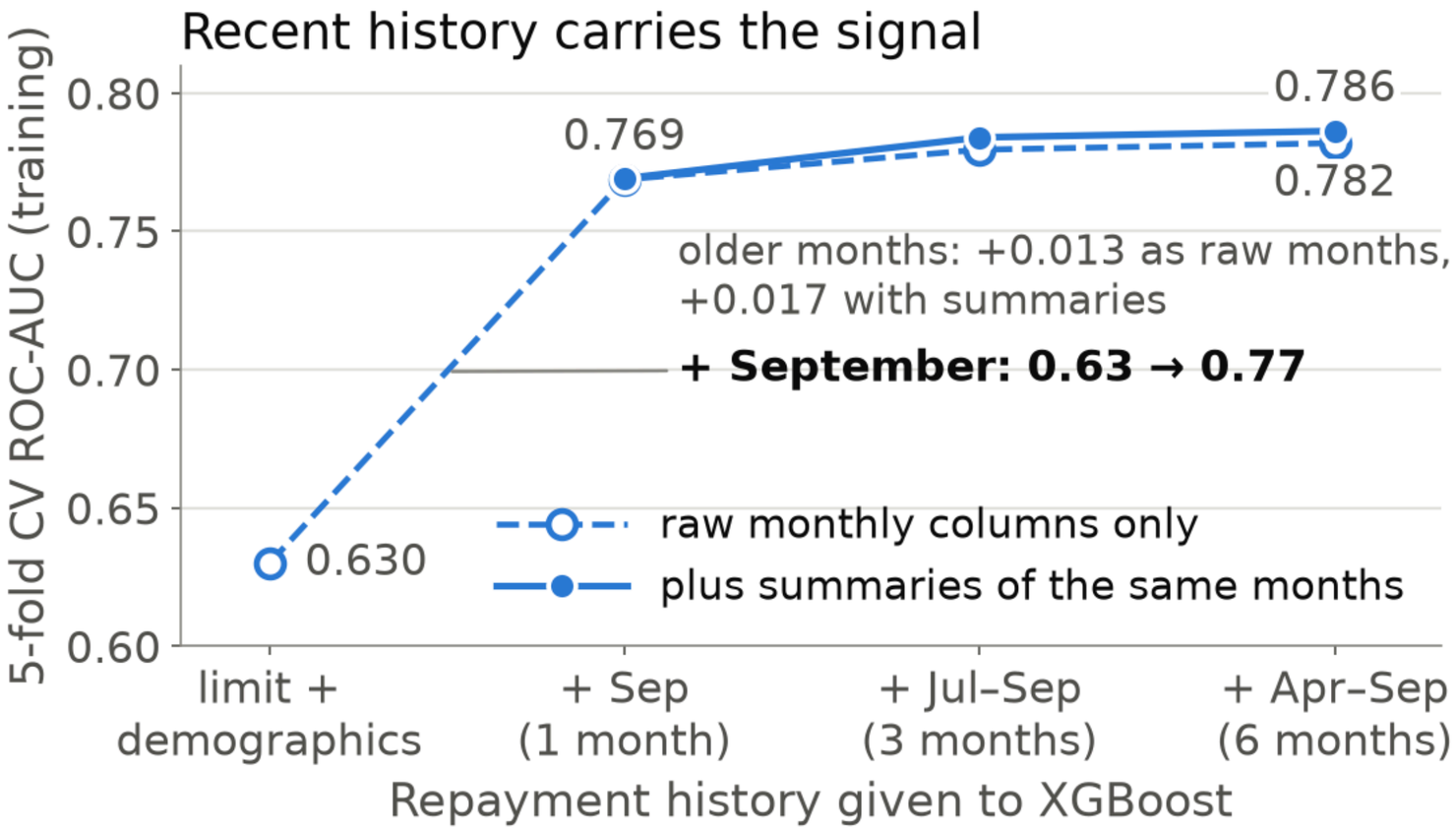


CV AUC by tree depth, rounds re-tuned; depth 1 means no interactions

Post-freeze training CV, never test-scored; results/ablation_depth.csv, pdp_summary.json; Friedman (2001)

Last month’s record carries most of the signal

September’s record lifts XGBoost from 0.63 to 0.77; older months add 0.017 (training CV).



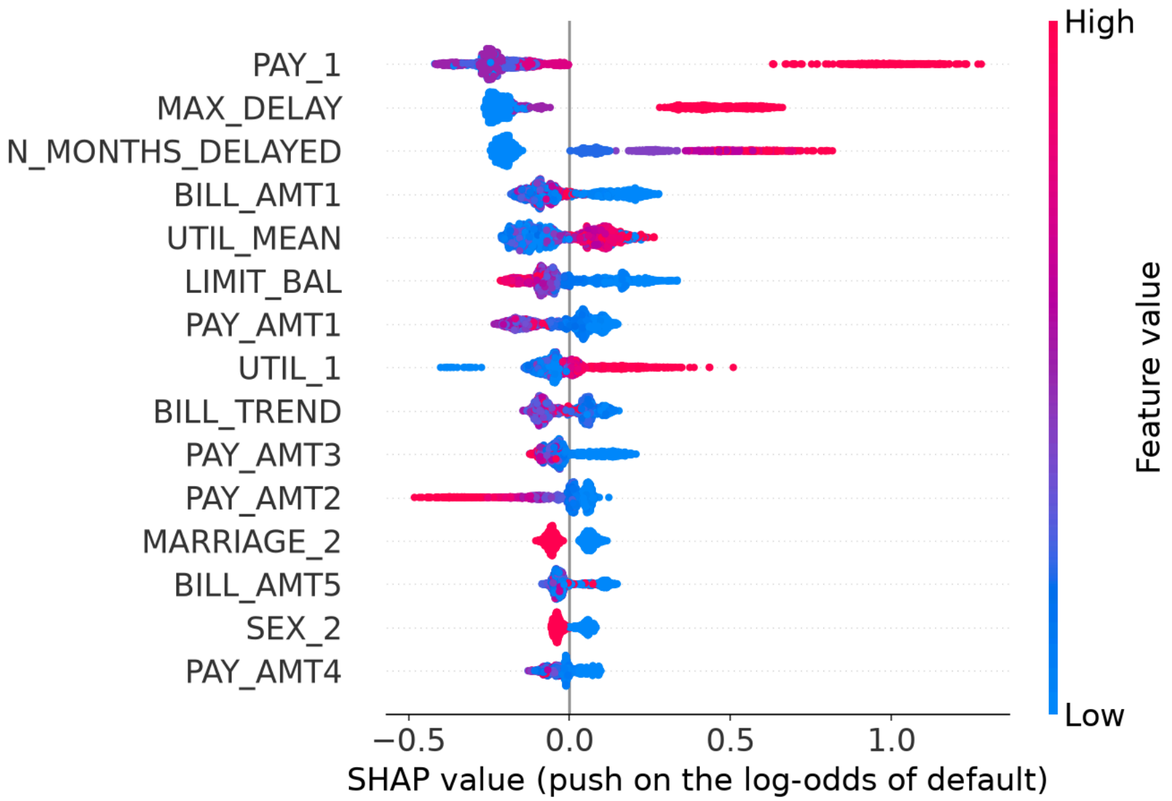
CV AUC as repayment history is added: raw months only (dashed, open circles) vs plus summaries of the same months (solid)

Post-freeze training CV, never test-scored; results/ablation_history.csv, pay_code_default_rates.csv (roll rates)

Arrears push scores up; a larger limit pulls down

SHAP splits each score into input contributions; September status ranks first on every measure.

TreeSHAP, tuned XGBoost: 2,000 training clients



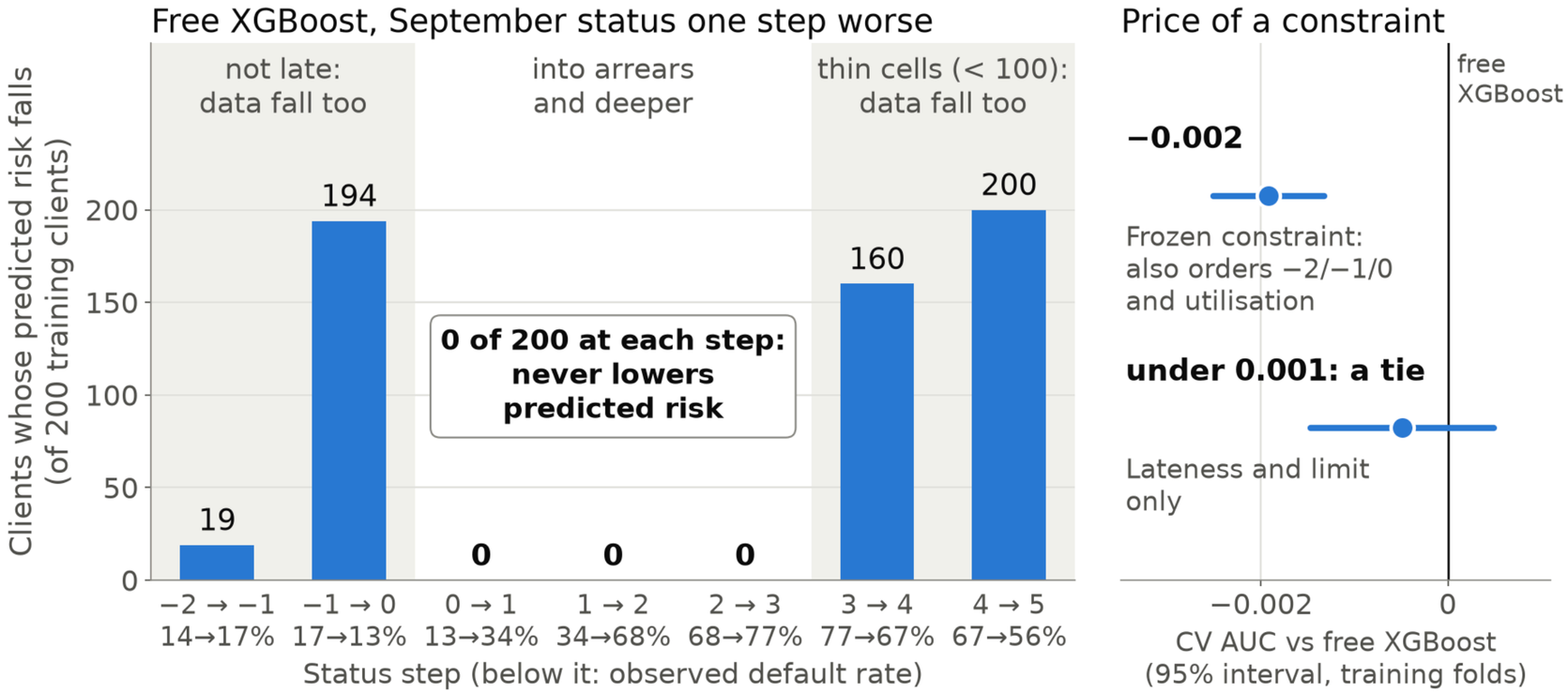
TreeSHAP for XGBoost, 2,000 training clients; red = high value.
PAY_1 = September status
MAX_DELAY = worst delay in six months
N_MONTHS_DELAYED = months in arrears
(these three carry one signal)
BILL_AMT1 / PAY_AMT1 = September bill / payment
UTIL = bill ÷ limit
LIMIT_BAL = credit limit
MARRIAGE_2 = single
SEX_2 = female

Training rows only; results/shap_importance.csv, shap_summary.json, rf_refit_drop_column.csv; Lundberg & Lee (2017); MAS FEAT (2018) Principle 13, https://www.mas.gov.sg/publications/monographs-or-information-paper/2018/feat; BoE & FCA (2024) §2.7, https://www.bankofengland.co.uk/report/2024/artificial-intelligence-in-uk-financial-services-2024

Monotone constraints: promise only what credit logic guarantees

Frozen version also orders −2/−1/0: costs 0.002 CV AUC; lateness and limit only: almost free.

Free XGBoost already respects lateness; forcing an order on −2/−1/0 costs AUC



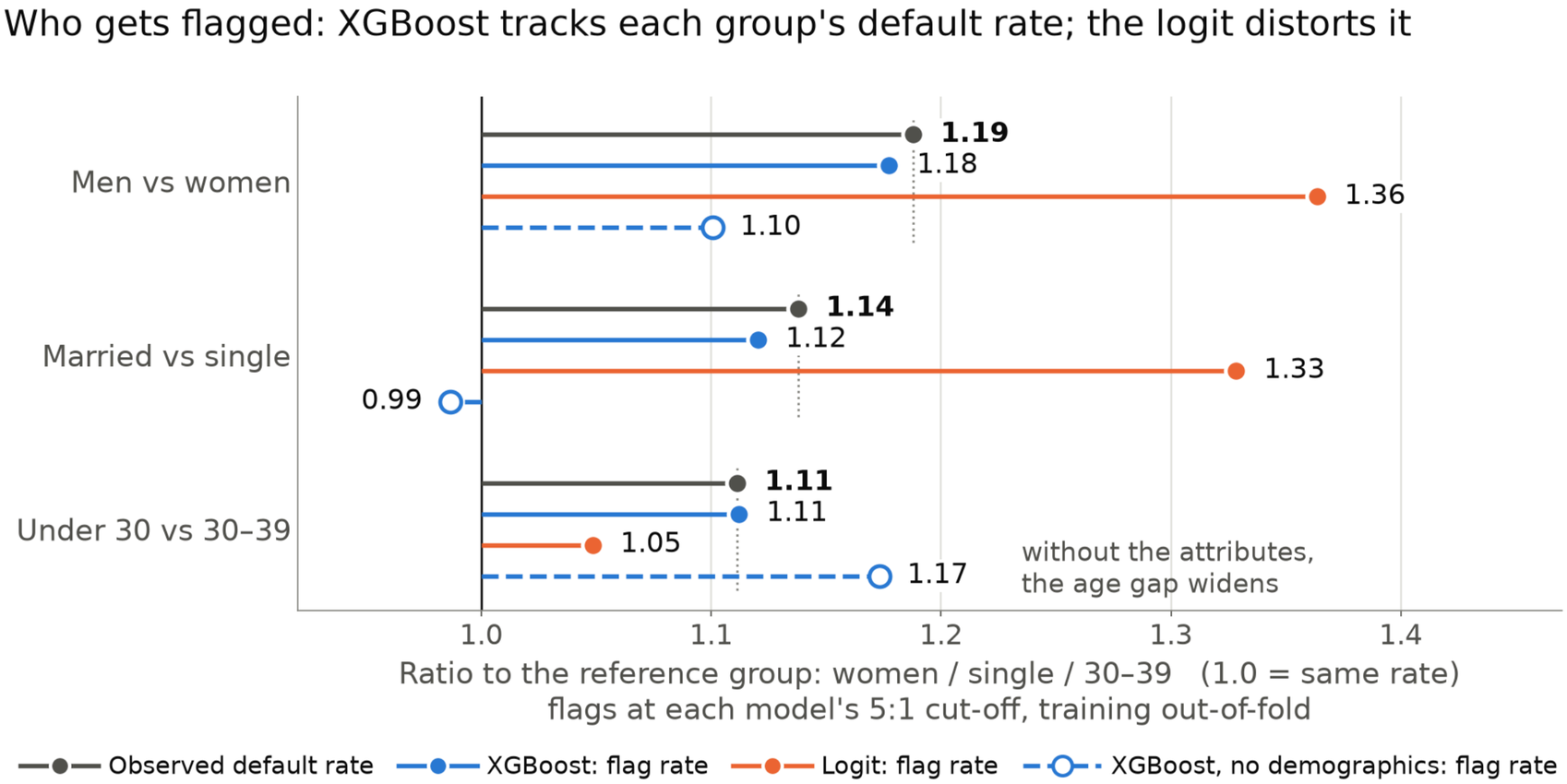
Left: 200 training clients, one status step worse. Right: CV AUC price of each constraint

Post-freeze training CV and training-row sweeps, never test-scored; results/monotone_check_consistent.csv (consistent sweep, 200 training clients), monotone_respec_cv.json; Chen & Guestrin (2016)

Personal attributes add little, but force a fairness choice

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Dropping sex, age, marriage, education costs 0.002 CV AUC; no option is fair on every measure.



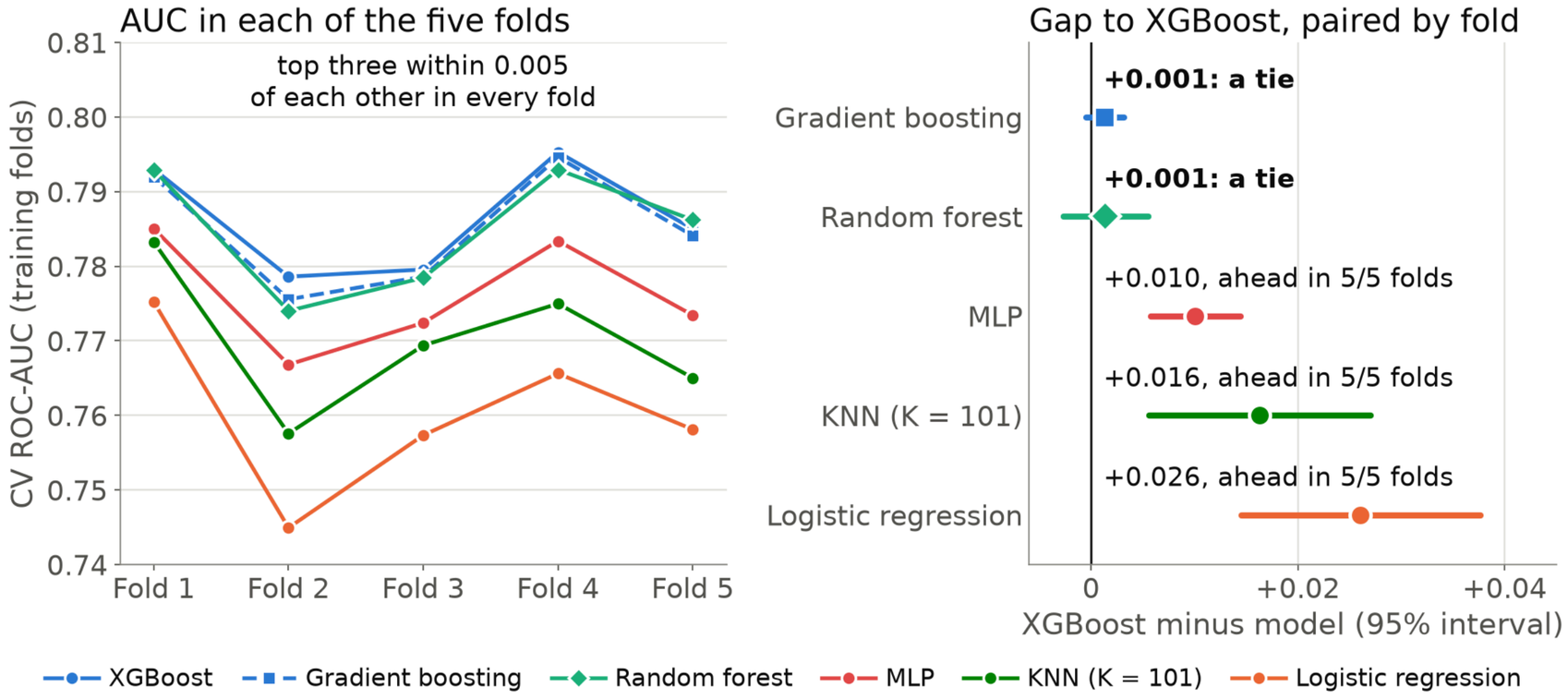
Flag-rate and default-rate ratios to the reference group at the cost-based cut-off (5:1 scenario, slide 18), training out-of-fold

Post-freeze training-OOF diagnostic, not a compliance assessment; results/fairness_groups.csv, fairness_summary.json, ablation_demographics.csv; EU AI Act Annex III 5(b), https://eur-lex.europa.eu/eli/reg/2024/1689/oj; Kleinberg et al. (2017)

Four tiers hold in every fold; top three tie

Ensembles, network, KNN, then the logit, in every fold; XGBoost's frozen test AUC is 0.789.

The three tree ensembles tie in every fold; their lead over KNN and the logit holds



Per-fold training CV AUC (left) and fold-paired gap to XGBoost (right)

Training OOF of the frozen configurations; results/fold_stability.csv, fold_stability_summary.json; test (n = 8,990, reporting only): aggregate_summary.json, auc_bootstrap_ci.csv

Four more families: close in ranking, not in probabilities

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Network 4th in every fold; LDA and naive Bayes too extreme (calibration slopes 0.749, 0.141).

	MODEL	HOW IT LEARNS	CV AUC	CALIBRATION SLOPE (1 = honest)
01	**Neural network (64–32)**	layers of curved boundaries, stopped early	**0.776**	**0.936**
02	**SVM (RBF kernel)**	widest margin; Platt turns scores into probabilities	**0.764†**	**0.997**
03	**LDA / naive Bayes**	bell-shaped amounts / independent bills: both false here	**0.757 / 0.736**	**0.749 / 0.141**

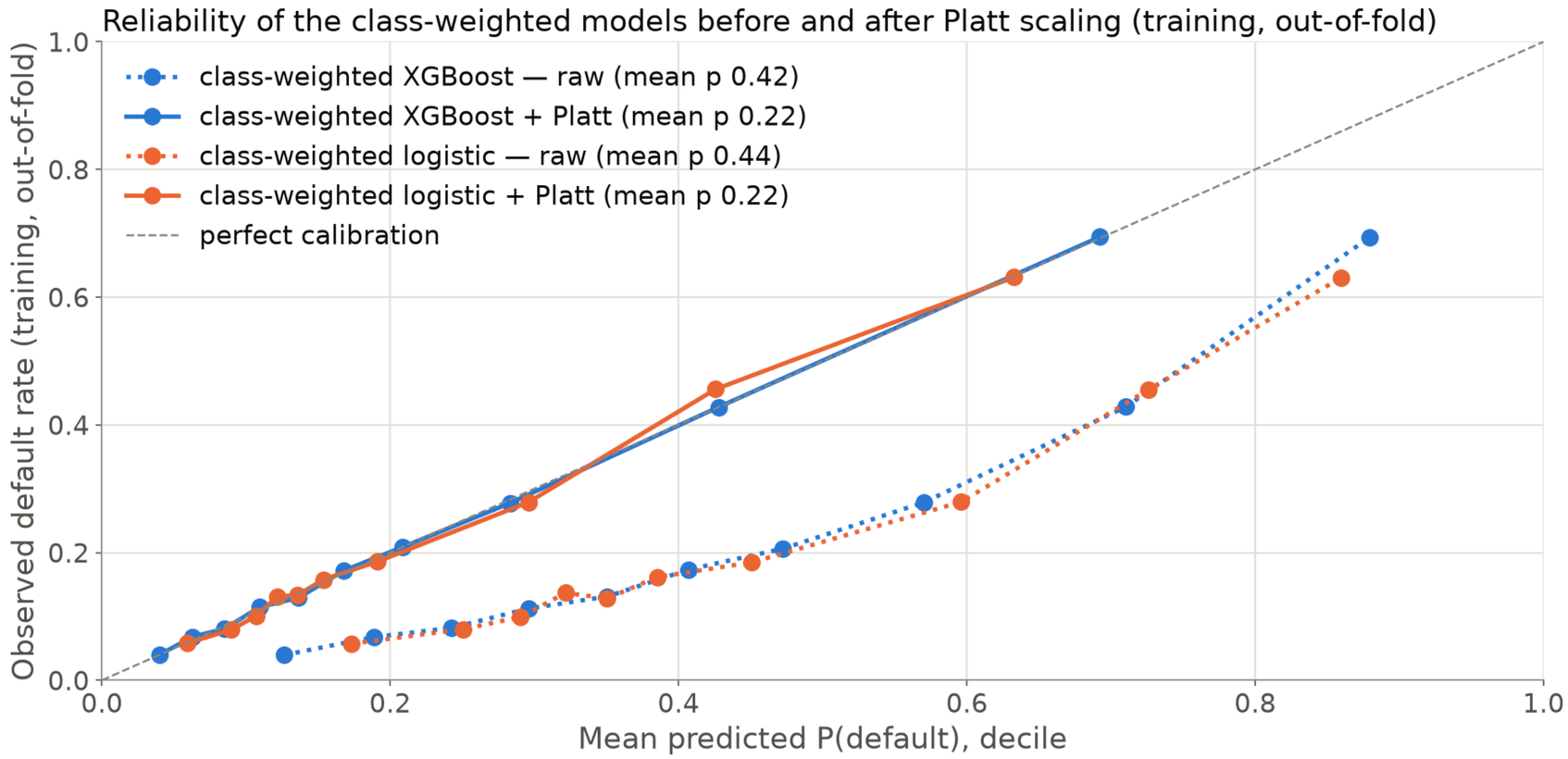
† SVM: mean of shared-fold OOF AUCs (tuned on 6,000 rows)

Training CV / OOF; results/fold_stability.csv (calibration slopes, post-freeze), mlp.json, svm_rbf.json, svm_selection_statistics.json, lda.json, naive_bayes.json

Class imbalance: weight the decision, not the data

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Weights move the intercept, not the ranking; weighted logit + Platt costs 0.4% more than plain.

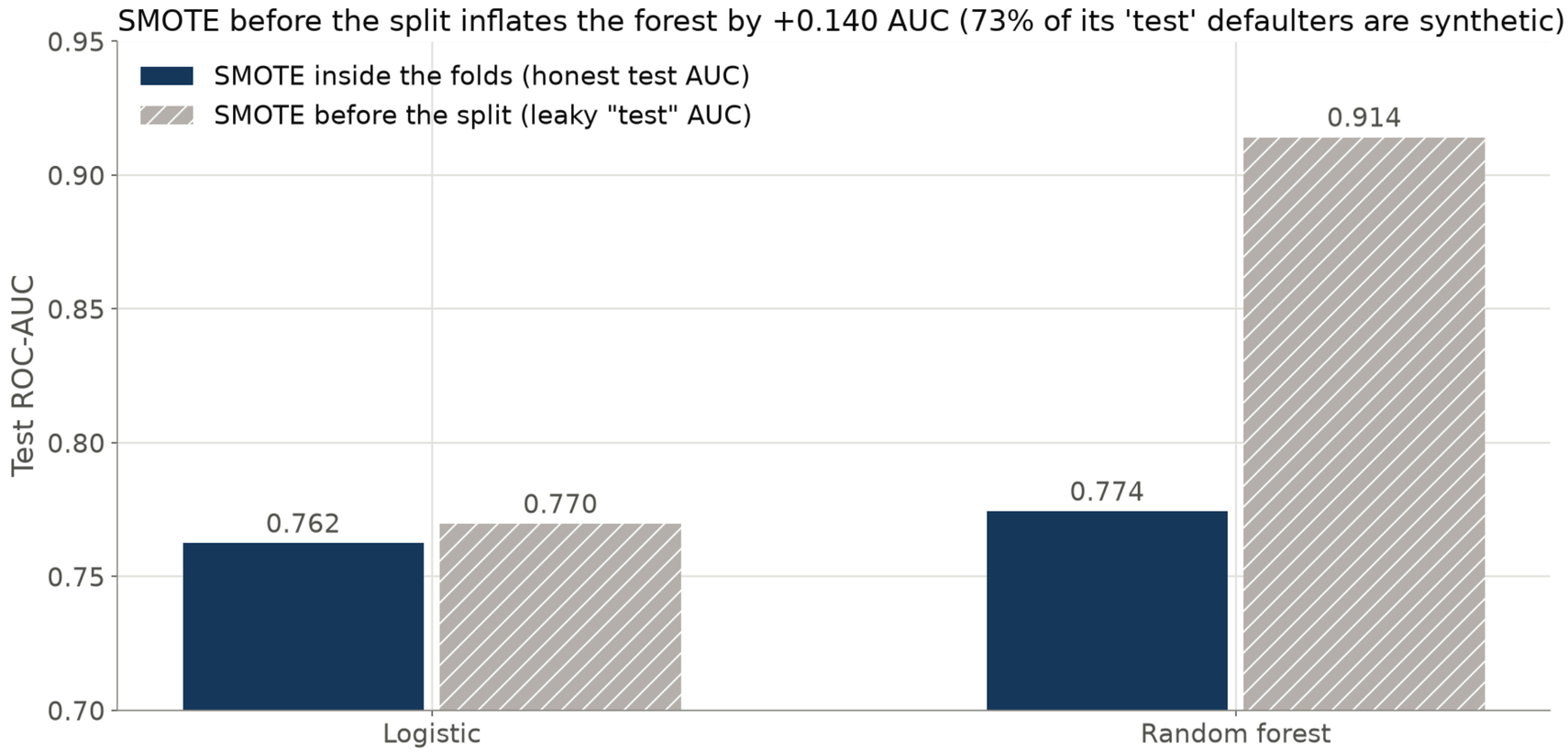


Weighted models before (dotted) and after Platt scaling (solid), training out-of-fold

Training OOF / CV; results/recalibration.csv, imbalance_comparison.csv, nested_cv_summary_repro.json (cross-fitted 5:1 cost +0.4%); Platt (1999); Chawla et al. (2002)

Leakage control: SMOTE before the split fakes 0.914 AUC

Honest forest 0.774; the same forest “tested” on 73% synthetic defaulters scores 0.914.

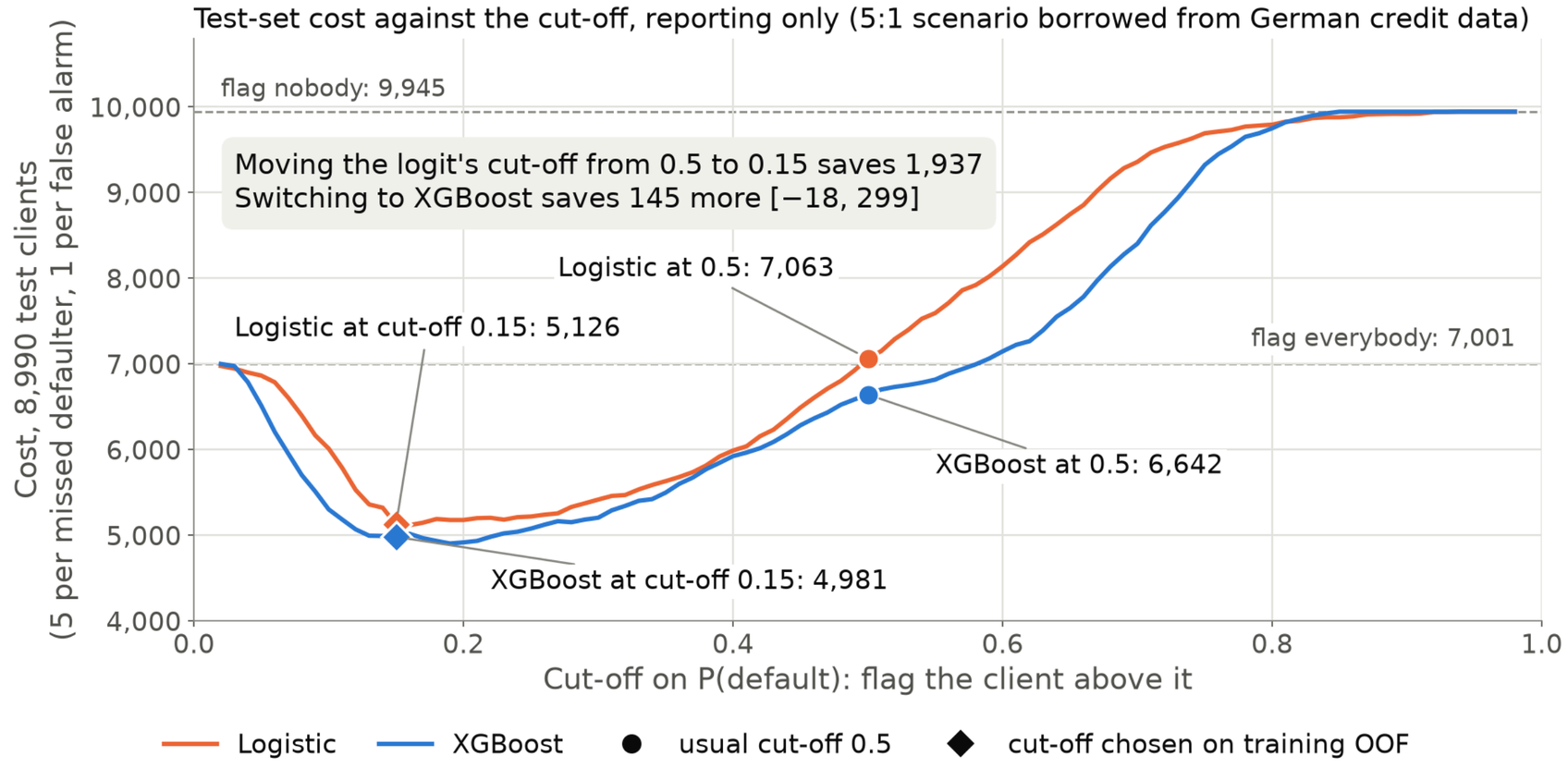


Same learners, same data: only the order of SMOTE and split differs

Test AUCs, reporting only (honest = frozen SMOTE-in-folds forest, CV also 0.774; leaky = “test” split of SMOTE’d data, n = 14,001); results/imbalance_leakage_demo.csv; negative control imb_leaky_smote_demo in frozen_manifest.json

The cut-off matters more than the model

At 5:1, moving the logit's cut-off saves 1,937; switching to XGBoost saves 145 more [−18, 299].

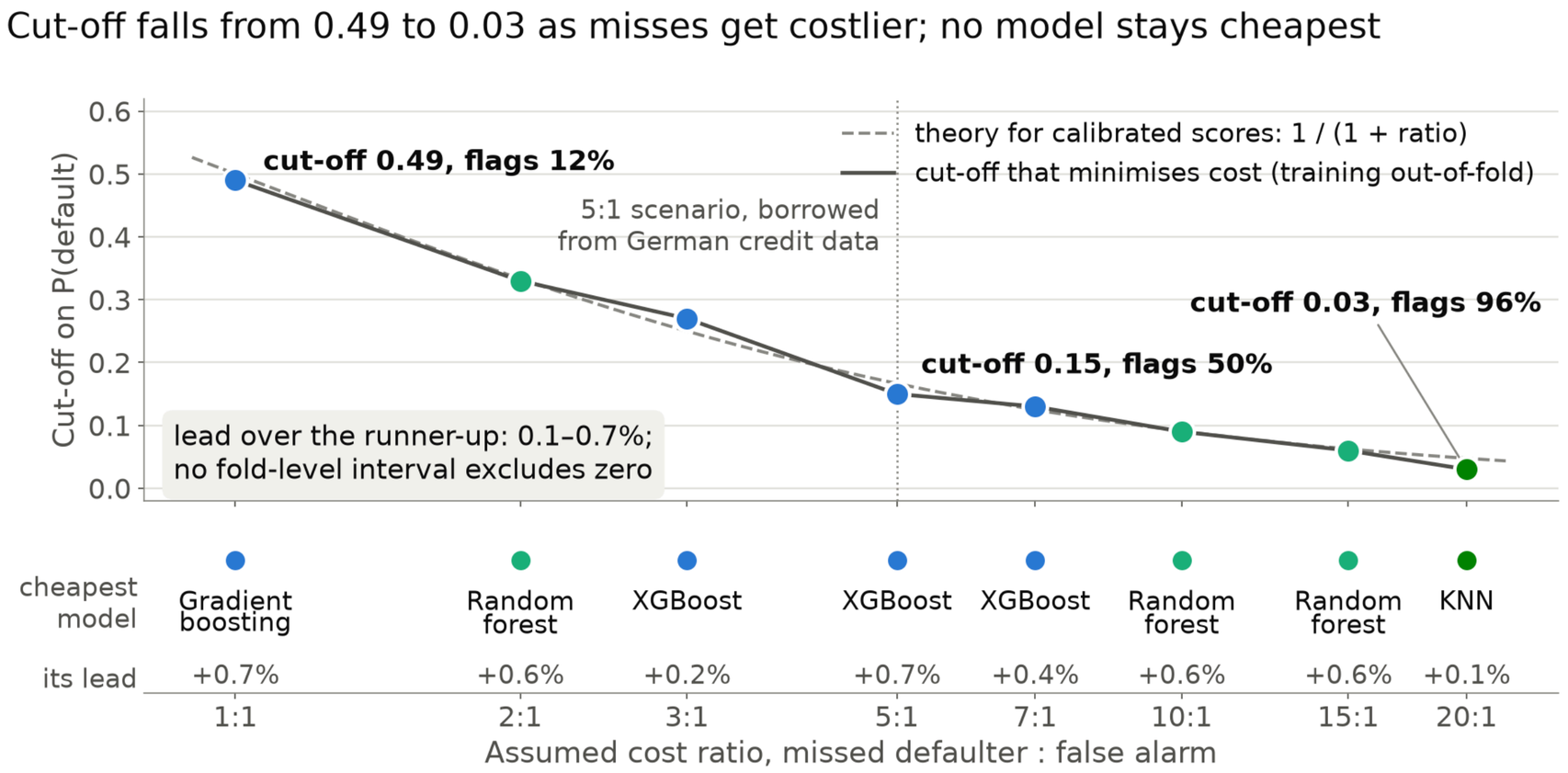


Test cost at 5:1 against the cut-off; diamonds = cut-offs chosen on training OOF

Cut-offs chosen on training OOF; test n = 8,990, reporting only; results/threshold_vs_model_saving.csv (paired bootstrap B = 1,000); 5:1 = Statlog German Credit cost matrix (Hofmann 1994), doi:10.24432/C5NC77

The bank's own cost ratio should set the cut-off

Cut-off falls from 0.49 at 1:1 to 0.03 at 20:1; the cheapest model keeps changing.

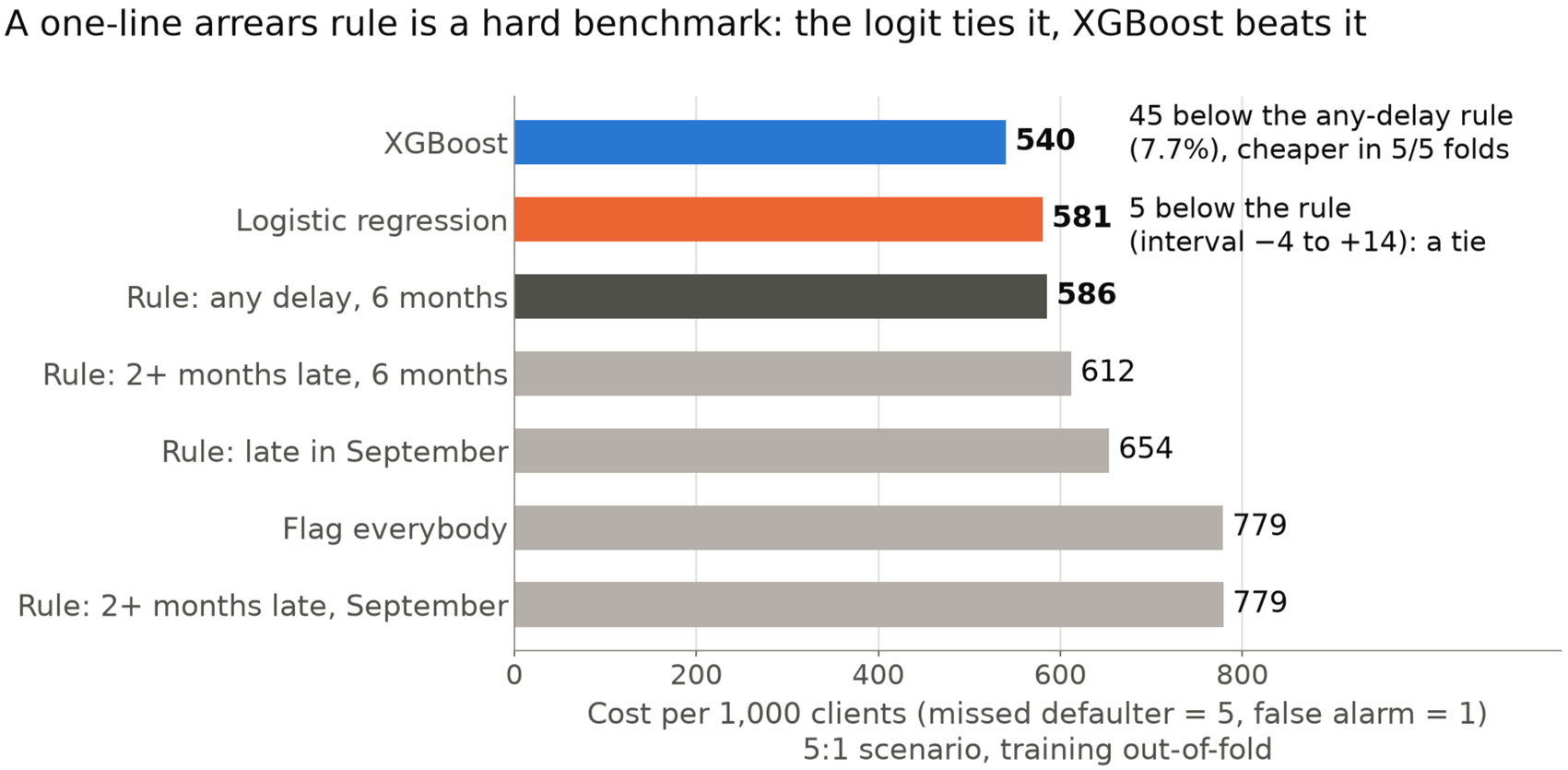


Training OOF: cost-minimising cut-off and cheapest model at each cost ratio

Training OOF, post-freeze, selects nothing; results/preferred_model_by_ratio_oof.csv, preferred_model_by_ratio_oof_long.csv, cost_anchors.json

A one-line rule gets most of the way

XGBoost costs 7.7% less than “flag any delay” at 5:1; the logit, almost nothing less.



5:1 cost per 1,000 training clients: no-parameter rules vs cross-fitted models

Training rows, post-freeze, never test-scored; results/simple_rule_benchmark.csv, topk_capture_oof.csv

Limits: one crisis vintage, a re-read test set, borrowed costs

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Rank with any ensemble, set the cut-off from costs, re-base before reuse.

01

VINTAGE

One bank, crisis onset: re-base probabilities, test a later period

02

TEST RECORD

Test read more than once, every read dated; one split; ensembles not nested

03

COSTS

5:1 borrowed: measure the bank's own losses, then reset the cut-off

Report §7; results/TEST_SET_LOG.md; Federal Reserve, charge-off and delinquency rates, https://www.federalreserve.gov/releases/chargeoff/delallsa.htm

Appendix · Nested CV, Platt, Cost Sweep, Settings and References

NESTED CV (RUAN, 30 Sep; reproduced 21 Oct on training rows only) — outer = the five project folds, inner 5-fold grid
KNN K = 101 uniform: nested AUC 0.768 (sd 0.007) → frozen choice confirmed · Logistic plain 0.760 (SE 0.005), 5:1 OOF cut-off 0.17, cross-fitted cost 12,195 → kept by the 1-SE rule · ridge 0.760 / lasso 0.760 within 1 SE · class-weighted 0.761, mean p 0.44 (mis-calibrated) · weighted + Platt 0.761, cross-fitted 5:1 cost +0.4% (12,247 vs 12,195; rule needs > 1% saving) → not adopted · nested CV run for KNN and the logistic family only; ensembles not nested

PLATT RECALIBRATION (19 Oct, nine weighted or resampled models) — maps fitted on training out-of-fold scores, thresholds cross-fitted; Brier returns to the unweighted twins (weighted XGBoost 0.177 → 0.131); 5:1 cut-offs 0.14–0.19; each calibrated version frozen at its first test scoring (19 Oct); maps fitted on training OOF only

COST SWEEP (threshold saving vs model saving, paired bootstrap B = 1,000 on the 8,990 test rows) — 1:1: 0 vs 87 [44, 131] · 3:1: 684 [541, 830] vs 127 [47, 214] · 5:1: 1,937 [1,604, 2,218] vs 145 [−18, 299] · 10:1: 7,121 [6,395, 7,791] vs 420 [187, 646] · 20:1: 20,335 [18,909, 21,621] vs 150 [−52, 333], cut-off 0.01 flags 99.9%

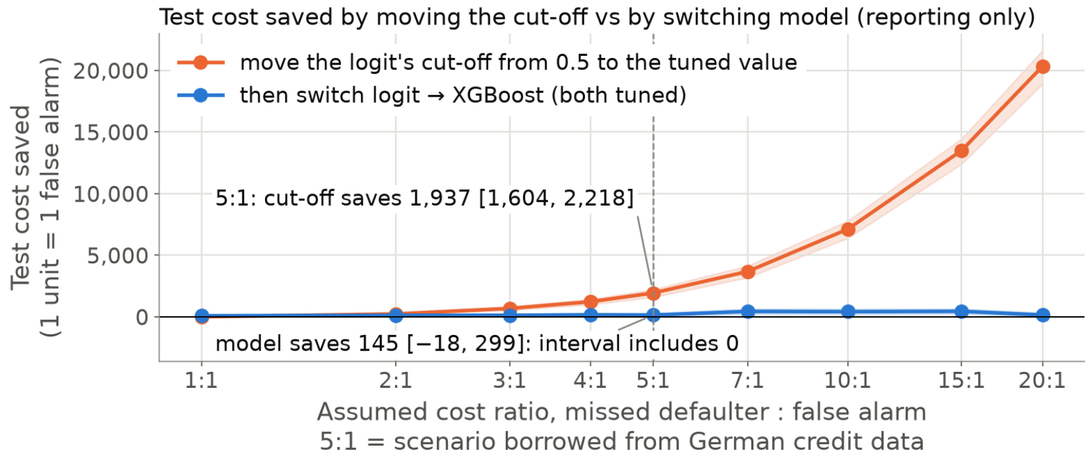
TUNED SETTINGS (5-fold CV AUC on 20,975 training rows, seed 2607526) — XGBoost depth 3, learning rate 0.03, 300 rounds · Random forest 500 trees, ≥ 20 clients per leaf, sqrt features · Pruned tree 17 leaves (cost-complexity path) · KNN K = 101 uniform · Logistic plain; L1/L2 at their CV optimum keep all 32 columns · RBF SVM C = 1, γ = 0.01, balanced weights (6,000-row tuning subsample) · MLP 32-64-32-1 with early stopping

POST-FREEZE CHECKS (21 Oct; training CV only, never test-scored, select nothing): pay-code recoding, interaction depth, history window, demographics and fairness, monotone respecification, simple rules, cost-ratio preference, fold stability; R refits match 13 of 15 Python CV AUCs within 0.005 — results/POST_REVIEW_CHECKS.md

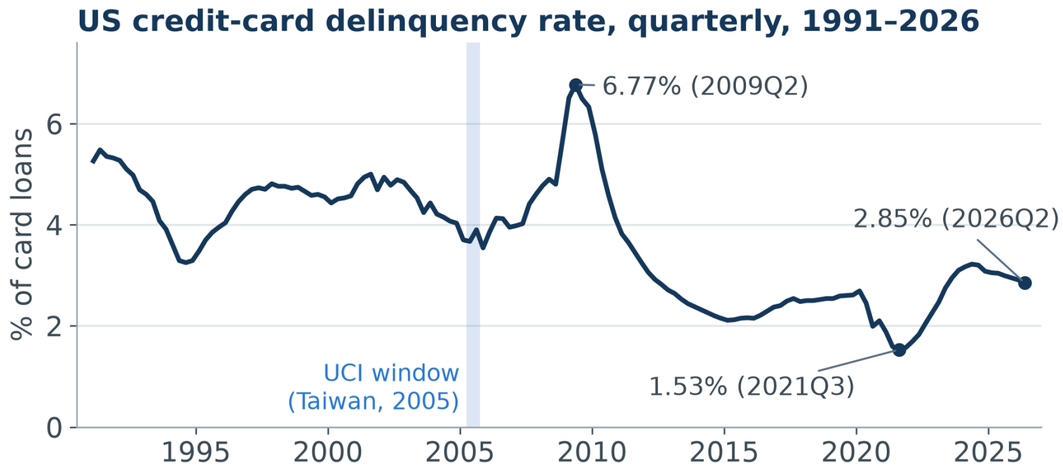
REFERENCES — Yeh, I.-C. & Lien, C.-H. (2009). Expert Systems with Applications 36(2) · James, Witten, Hastie & Tibshirani, An Introduction to Statistical Learning · Breiman, L. (2001). Random forests. Machine Learning 45 · Chen, T. & Guestrin, C. (2016). XGBoost. KDD · Chawla, N. V. et al. (2002). SMOTE. JAIR 16 · Lundberg, S. & Lee, S.-I. (2017). A unified approach to interpreting model predictions. NeurIPS · Platt, J. (1999). Probabilistic outputs for support vector machines · Hofmann, H. (1994). Statlog German Credit, UCI · Pedregosa, F. et al. (2011). Scikit-learn. JMLR 12 · Friedman (2001) · Nadeau & Bengio (2003) · Kleinberg, Mullainathan & Raghavan (2017) · EU AI Act 2024/1689 Annex III 5(b) · BoE & FCA (2024) · FSC Banking Bureau index-7.xlsx · CBC FSR Jun 2008 · Code, data and results: github.com/jingwen013-design/mh6805-credit-default

results/nested_cv_summary_repro.json, recalibration.csv, threshold_vs_model_saving.csv, POST_REVIEW_CHECKS.md, r_crosscheck.csv

Appendix · Cost sweep and real-world context



Test set, reporting only; results/threshold_vs_model_saving.csv



Federal Reserve, https://www.federalreserve.gov/releases/chargeoff/delallsa.htm (US analogy, 30+ days past due)

Official Journal of the European Union 2024/1689 12.7.2024

ANNEX III

High-risk AI systems referred to in Article 6(2)

5. Access to and enjoyment of essential private services and essential public services and benefits:

(b) AI systems intended to be used to evaluate the creditworthiness of natural persons or establish their credit score, with the exception of AI systems used for the purpose of detecting financial fraud;

Excerpt; blue box added. Regulation (EU) 2024/1689 (AI Act), Annex III 5(b), OJ L, 12.7.2024 · eur-lex.europa.eu/eli/reg/2024/1689/oj

Regulation (EU) 2024/1689 Annex III 5(b), https://eur-lex.europa.eu/eli/reg/2024/1689/oj (application of Annex III obligations moved to 2 Dec 2027 by Regulation (EU) 2026/1744)

2.7: Explainability

A high proportion of firms currently using AI (81%) employ some kind of explainability method.

Feature importance (72%) and Shapley additive explanations (64%) are the most popular explainability approaches.

BoE & FCA, AI in UK financial services 2024, §2.6–2.7 (excerpt of §2.7; blue box added), https://www.bankofengland.co.uk/report/2024/artificial-intelligence-in-uk-financial-services-2024